

Integration of Multivariate Data Streams With Bandpower Signals

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Abstract—The urge to further our understanding of multimodal neural data has recently become an important topic due to the ever increasing availability of simultaneously recorded data from different neural imaging modalities. In case where EEG is one of the modalities, it is of interest to relate a nonlinear function of the raw EEG time-domain signal, say, EEG band power, to another modality such as the hemodynamic response, as measured with NIRS or fMRI. In this work we tackle exactly this problem defining a novel algorithm that we denote multimodal source power correlation analysis (mSPoC). The validity and high performance of the mSPoC framework is demonstrated for simulated and real-world multimodal data.

Index Terms—EEG, EEG-NIRS, multimodal, neuroimaging, NIRS.

I. INTRODUCTION

MULTIMODAL neuroimaging has become an important tool for basic research and clinical diagnosis [1], [2]. A very popular multimodal setup is the combination of non-invasive electroencephalogram (EEG) [3] data and hemodynamic measurements, obtained by functional magnetic resonance imaging (fMRI) [4] or near-infrared spectroscopy (NIRS) [5]. The technical challenges associated with the recording of these combined measurements have been successfully mastered

during the last two decades. Nonetheless there is still no gold standard for analyzing multimodal neuroimaging data [1].

This article is organized as follows. In Section II we recapitulate a widely accepted generative model that connects measures of neural oscillations (EEG/MEG) with measures of hemodynamic responses (fMRI/NIRS). This sets the stage for Section III, in which we review state-of-the-art methods that aim to analyze such multimodal data. Most of these methods either do not use the full information available or violate assumptions of the generative model. This leads to the presentation of a novel analysis technique for multimodal measurements in Section V, which for the first time incorporates all relevant aspects of the data model. Sections VI and VII are devoted to the application of our method to simulated and real multimodal imaging data. We conclude our paper with a discussion in Section VIII.

II. GENERATIVE MODEL FOR MACROSCOPIC EEG AND HEMODYNAMIC SIGNAL

Let us first introduce a physiologically motivated generative model for the simultaneous measurement of EEG and the hemodynamic signal. We denote the EEG signal by a vector-valued time series $\mathbf{x}(t) \in \mathbb{R}^{N_x}$ and the hemodynamic measurements by $\mathbf{y}(t) \in \mathbb{R}^{N_y}$, where N_x and N_y are the number of sensors of the EEG measurements and the hemodynamic data, respectively. In case the hemodynamic measurements are obtained by fMRI, N_y is the number of voxels and can be very large. Both signals depend on the hidden neural activity, which can be written as a multivariate time series $\mathbf{s}(t) \in \mathbb{R}^{N_s}$, where each component denotes the time-dependent activation in one of N_s brain regions.

The neural activation of a brain region is often described as electrical dipoles associated with the synchronized activity of *neural populations*. The resulting electrical fields are directly measured by the EEG electrode on the scalp. However, since each EEG electrode does not capture the field of a single brain region, but always a linear and instantaneous superposition of all the individual fields (see e.g., [6] or [7]), we can model the measured EEG as

$$\mathbf{x}(t) = \mathbf{A}_x \mathbf{s}(t) + \epsilon_x(t) \quad (1)$$

where $\mathbf{A}_x \in \mathbb{R}^{N_x \times N_s}$ is a linear mapping from the space of brain regions to the space of electrodes and $\epsilon_x(t)$ a noise term that accounts for all non-neural activity. The linear relationship relies on the assumption that magnetic induction and capacitive effects do not influence the measurements.

In the case of hemodynamic recordings the relationship between neural dipoles and measured signal is more complicated

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[8]. The hemodynamic signal reflects the energy demand induced by neural activity. Neurons consume most energy for restoration of the resting membrane potential after neural signals from other cells depolarized a cell [9]. Let us denote the neuronal signal at some brain region i with $s_i(t)$ and the local vasomotor feed forward signal (that gives rise to the hemodynamic signal) at that brain region with $p_{s_i}(t)$. Usually, $p_{s_i}(t)$ is modeled by the instantaneous variance of the neural dynamics, the band power in a certain frequency band or a more general nonlinear function of $s_i(t)$. Since the hemodynamic response is not instantaneous, the hemodynamic signal is modeled by a convolution of the past of $p_{s_i}(t)$ (alternatively called causally filtered $p_{s_i}(t)$):

$$q_i(t) = \phi(s_i(t)) := (h \star p_{s_i})(t) = \sum_{\tau > 0} h(\tau) p_{s_i}(t - \tau) \quad (2)$$

where $\star$ denotes convolution and the nonlinear convolution operator ϕ combines the energy consumption nonlinearity with a linear convolution $h(\tau)$, also known as *hemodynamic response function* (HRF). In a final step, these signals are linearly transformed into the sensor space of the measurement device (NIRS or fMRI):

$$\mathbf{y}(t) = \mathbf{A}_y \Phi(\mathbf{s}(t)) + \epsilon_y(t), \quad (3)$$

where $\mathbf{A}_y \in \mathbb{R}^{N_y \times N_s}$, $\epsilon_y(t)$ is modality-specific noise, and $\Phi(\mathbf{s}(t)) = (\phi(s_1(t)), \dots, \phi(s_{N_s}(t)))^\top$.

The electric fields produced by cortical sources indeed superimpose linearly and instantaneously at the EEG electrodes, which means that the mixing model for the EEG electrodes (eq.(1)) is exact. Note, however, that the model for the vascular signal (cf. Equation (3) and Fig. 1) contains an approximation. We assume that this model can be separated into two steps: the first step is a nonlinear mapping ϕ (here consisting of a band power transformation followed by a canonical HRF) that is applied to each of the cortical sources independently. This nonlinearity also contains all the dependencies between different time instances. The second step mixes the intermediate signals $\mathbf{q}(t) = \phi(\mathbf{s}(t))$ instantaneously to obtain the measured vascular signals. In other words: nonlinearity and mixing in time is separable from the spatial mixing. This is not necessarily the case; different brain regions might have different HRF profiles. However, this separability assumption is usually true to a very good approximation and widely used in the neuroscience literature [10].¹

We further assume that we know the nonlinearity/temporal mixing ϕ . The goal of the multimodal data integration problem is then to estimate the unknown spatial mixing matrices $\mathbf{A}_x$ and $\mathbf{A}_y$ to establish the relationship between surface EEG and measured vascular signals (i.e., through NIRS or fMRI).

¹Note, that our model does not rely on a *strict* separability assumption; different neural sources are indeed allowed to have different HRFs, only the mapping (mixing) of each source into different voxels or optode signals is fixed. Effectively this assumes *local* space-time separability within a brain region that constitutes a dipole, yet allows *global* space-time non-separability across the brain.

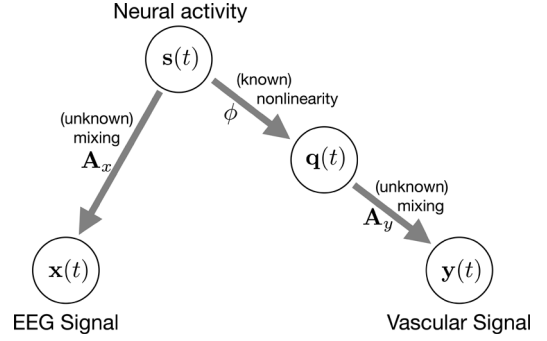


Fig. 1. Generative model of multimodal EEG and hemodynamic signal corresponding to (1) and (3) (noise terms omitted).

III. STATE OF THE ART IN MULTIMODAL NEUROIMAGING

Many methods have been proposed to solve the data integration problem underlying (1) and (3), for a review see e.g., [1]. Most data integration approaches can be categorized into model-driven approaches, supervised methods or unsupervised techniques.

Model driven approaches are based on biophysically inspired forward models. A good model is the more important the further away from the neural generators a measurement is. Hence especially non-invasive multimodal studies using EEG and hemodynamic signals use model based analyses [11]–[13]. Sometimes, however, model based approaches can be difficult to validate; in particular is not clear which level of detail is optimal for modeling the biophysical processes underlying (1) and (3). Many models use a neural mass model for the net metabolic cost of neural activity to hemodynamic signals [14]; this quantity is very similar to band power signals.

Multimodal supervised analyses use brain activity measurements to predict a stimulus variable. As multimodal measurements often yield complementary views on brain activity they can often predict stimulus information better than unimodal measurements alone [15]. But not in all experimental paradigms a stimulus variable is available; examples are resting state analyses or detection of epileptic activity, for a review see [16]. A solution is to use one modality to predict the other modality as e.g., in [17], [18]. Similarly one can use information from one modality to constrain the solution found by another modality [19]–[22]. In these *asymmetric supervised analyses* one has to decide which modality will serve as regressor or more generally how much weight the features from one modality should obtain as opposed to another modality in the analysis.

When neither forward model nor stimulus information is available unsupervised methods can be used to analyze data in an explorative way. Popular examples are principal component analysis (PCA) [23], [24], independent component analysis (ICA) [25]–[27] and canonical correlation analysis (CCA) [28]. In multimodal settings, unsupervised methods have become indispensable for artifact removal [29] as well as for analysis. Both PCA and ICA based analyses are inherently unimodal methods and are typically computed on data from one modality only, before they are combined in a second step. A number of solutions to these problems have been proposed, see e.g.,

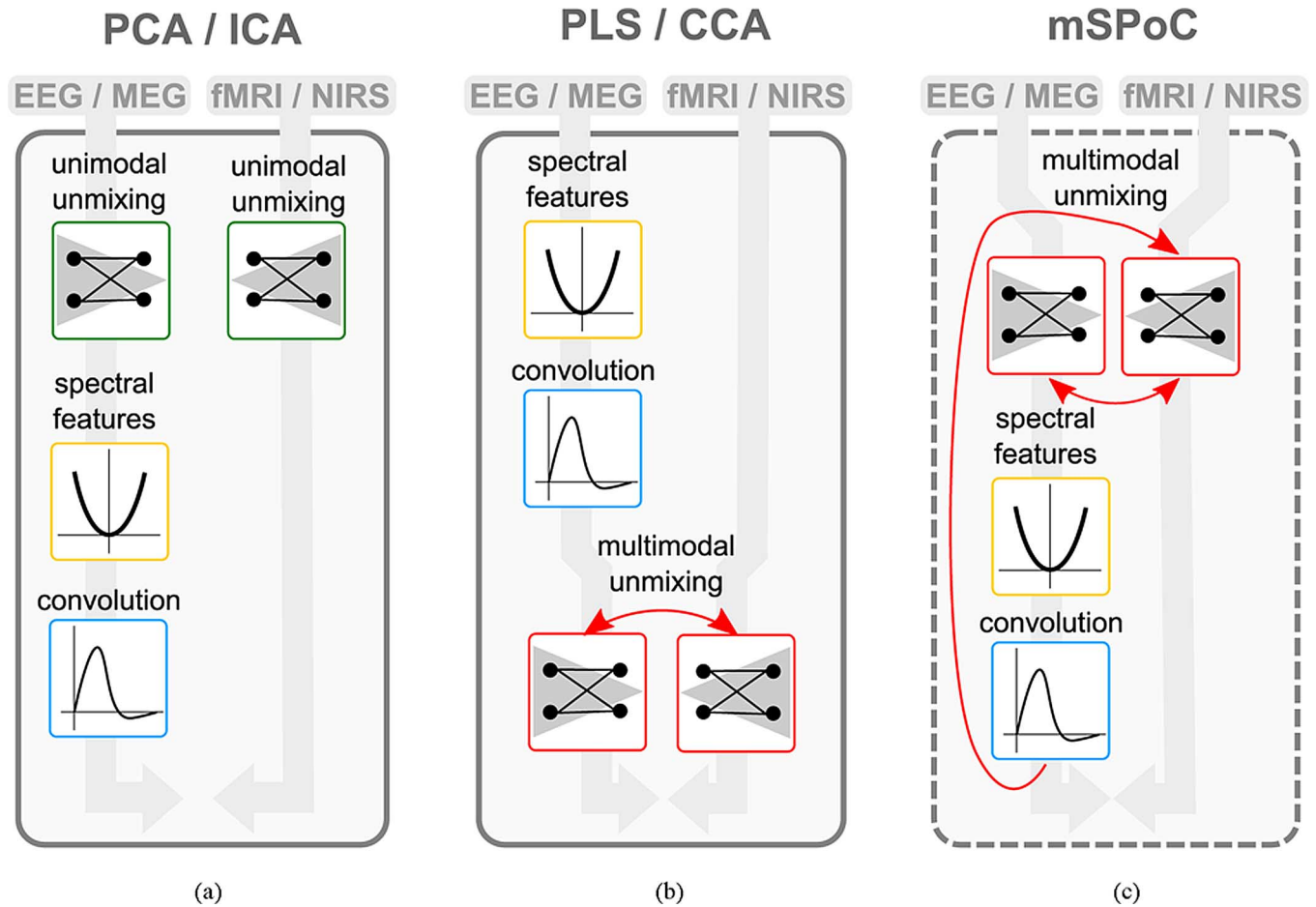


Fig. 2. Schematic comparison of two popular unsupervised analysis approaches to multimodal neuroimaging data, as well as our own novel approach. The three processing streams are to be read from top to bottom, i.e., multivariate measures of neural oscillations (EEG/MEG) and of hemodynamic responses (fMRI/NIRS) come in at the top and are then subjected to several processing steps, such as an unmixing, the estimation of spectral features, and convolution with an HRF-like function. *Left*: PCA/ICA based approaches usually perform a separate unmixing before bandpower estimation and convolution. The unmixing stage is uninformed of the respective other modality. *Middle*: In approaches based on PLS/CCA there is bandpower extraction and convolution as preprocessing steps before the unmixing state. Here the unmixing stage acknowledges the inherent multimodality of the data, yet the nonlinear preprocessing may hinder the linear unmixing. See Sections III and IV for more details. *Right*: In this paper we propose an (iterative) approach to the unmixing that is multimodal and takes the nonlinear features of the resulting components into account. See Section V for more details.

[30]–[34]. CCA (or equivalently partial least squares (PLS) [35]) does not suffer from this limitation. It is inherently multimodal and has become a successful tool for data integration. Some examples of CCA based approaches are [36]–[38]. A limitation of these integration approaches is that they are often based on band power signals. When working with invasively recorded signals as in e.g., [38] this is not a problem. However computing the band power on EEG surface electrode time series might lose important information, as explained in Section IV.

Multimodal neuroimaging of NIRS and EEG has witnessed an increased interest in recent years with studies appearing in various neuroscientific domains [15], [39]–[41], however the development of necessary multivariate tools for multiple sources has not caught up with this development to this date.

Taken together the state-of-the-art in multimodal data fusion of non-invasive brain signals are based on either detailed biophysical models, inherently unimodal approaches or band power signals computed in sensor space. Each of these strategies has limitations. Validation of biophysical models needs technically challenging physiological studies such as [42]–[44].

Inherently unimodal methods require trading of one modality against the other [21], [22] and different preprocessing parameters might lead to very different fusion results [45]. And finally the popular approach of analyzing band power time series computed from *single* electrodes is suboptimal, as explained in Section IV. Fig. 2 schematically compares two of the reviewed popular approaches to the analysis of multimodal neuroimaging data in the context of electrophysiological signals and hemodynamical recordings, namely PCA/ICA and PLS/CCA based approaches.

IV. LIMITATIONS OF BAND POWER REPRESENTATION

Inverting the generative model given in (1), (2), and (3) requires the inversion the mixing matrices $\mathbf{A}_x$ and $\mathbf{A}_y$ (i.e., unmixing) and the estimation of EEG band power. The computation of band power is a nonlinear operation. Thus, we would like to emphasize, the two processing steps (i) linear (un-)mixing and (ii) band power computation do not commute. In many approaches reviewed in the previous sections, however, the EEG band power is computed for each recording channel first and

thereafter a linear unmixing is carried out. The following paragraph illustrates why this course of action is not in line with the generative model.

First of all, we express the channel-wise EEG band power as a function of the assumed underlying sources and their corresponding spatial patterns, i.e., the vector $\mathbf{s}(t)$ and the matrix $\mathbf{A}_x$. Let us denote the multivariate and complex-valued time-frequency representation of the EEG time series by $F\{\mathbf{x}\}(t, f)$ and its spectral power by $P\{\mathbf{x}\}(t, f)$. From here on we will omit the time and frequency index for simplicity of notation. Since $\mathbf{x}(t)$ is a vector-valued quantity, we have $F\{\mathbf{x}\} \in \mathbb{C}^{N_x}$ and $P\{\mathbf{x}\} \in \mathbb{R}^{N_x}$, with the spectra being computed in each dimension of $\mathbf{x}$ separately. Furthermore, we have $P\{\mathbf{x}\} = \text{diag}(F\{\mathbf{x}\} F\{\mathbf{x}\}^\dagger)$, where $F\{\mathbf{x}\}^\dagger$ denotes the transposed complex conjugate of $F\{\mathbf{x}\}$ and $\text{diag}(\mathbf{M})$ denotes the diagonal of the square matrix $\mathbf{M}$. Then we have

$$\begin{aligned} F\{\mathbf{x}\} &= F\{\mathbf{A}_x \mathbf{s}\} \\ &= \mathbf{A}_x F\{\mathbf{s}\} \text{ and} \\ P\{\mathbf{x}\} &= \text{diag}(F\{\mathbf{x}\} F\{\mathbf{x}\}^\dagger) \\ &= \text{diag}(\underbrace{\mathbf{A}_x F\{\mathbf{s}\} F\{\mathbf{s}\}^\dagger \mathbf{A}_x^\top}_{:=\mathbf{C}}) \\ &= \text{diag}(\mathbf{C}), \end{aligned} \quad (4)$$

where $\mathbf{C}$ is the cross-spectrum matrix that depends on time and frequency (parameters omitted). From the considerations above, it follows that the spectral power at sensor k is given by

$$\begin{aligned} P\{\mathbf{x}\}_k &= \sum_i^{N_s} a_{k,i}^2 P\{s_i\} \\ &+ 2 \sum_{i,j;i < j}^{N_s} a_{k,i} a_{k,j} \Re(F\{s_i\} F\{s_j\}^\dagger), \end{aligned} \quad (5)$$

where $a_{k,i}$ denotes the element at row k and column i in the matrix $\mathbf{A}_x$ and $\Re(\cdot)$ denotes the real part of its complex number argument. The last equation shows that the spectral power computed at channel k is not just a linear superposition of the spectral powers of all neural sources whose activity reaches that channel. Instead, it is also biased by the cross-spectra between the sources. Thus, we would like to point out that a linear combination of channel-wise computed band power (i.e., $\mathbf{w}^\top P\{\mathbf{x}\}$) will in general not be able to recover what we are really interested in, namely the time course of the source band power (i.e., $P\{\mathbf{w}^\top \mathbf{x}\} = P\{\hat{s}\}$).

We further illustrate this point with a simple toy data example given in Fig. 3. In this example we focus only the EEG part of the multimodal problem. We assume that we have only two recording channels (denoted by x_1 and x_2), that the data from the other modality has been projected to a univariate time series (denoted by z), and that the coupling between the power of the EEG and the time course of the other modality is not time delayed. We simulated two sources (s_1 and s_2), such that the power time course of s_1 is correlated with z . Thereafter we linearly mixed the time courses of s_1 and s_2 by means of a rotation matrix, i.e., we mapped the s_1/s_2 coordinate system (source space) to the x_1/x_2 coordinate system (sensor space) simply by rotating the former. The scatter plot on the left side in the

figure shows the mixed raw data in sensor space, with the red line corresponding to the direction of s_1 . This direction represents the target source, because only the power time course of s_1 is coupled to z . The black line in the scatter plot corresponds to s_2 and can be regarded as a distractor. The scatter plot in the middle depicts the data after channel-wise band power computation (here approximated by the variance within consecutive time windows). The search space of the optimal weight vector $\mathbf{w}$ is only one dimensional in this example, because $\mathbf{w}$ can be parameterized by its angle with respect to one of the coordinate axis (here the angle with respect to the x_2 axis). In the plot on the right side, we show the covariance between the linear combination of channel-wise power (i.e., $\mathbf{w}^\top P\{\mathbf{x}\}$) and z as well as the covariance between the power time course of a linear combination of channels (i.e., $P\{\mathbf{w}^\top \mathbf{x}\}$) and z . Both covariances are shown as a function of the direction of $\mathbf{w}$. The direction corresponding to s_1 is indicated by a red vertical line in this graph. It can be seen that in this particular scenario the direction of maximal covariance between $\mathbf{w}^\top P\{\mathbf{x}\}$ and z does not coincide with the direction of the true source. In fact, the direction that is found when optimizing $\mathbf{w}^\top P\{\mathbf{x}\}$ is actually orthogonal to the true source direction. The reason that optimizing $\mathbf{w}^\top P\{\mathbf{x}\}$ is suboptimal here is that computing the nonlinearity before the spatial filtering step loses sign information. This in turn leads astray the search for the true source.

V. MULTIMODAL SOURCE POWER CORRELATION

In this section we propose a novel method that is inherently multimodal and specifically designed to invert the generative model given in (1), (2), and (3). Our method assumes a set of underlying sources whose activity is visible in both imaging modalities. The coupling between these sources is maximized and therefore we refer to our method as *multimodal source power correlation* (mSPoC). The objective function we will define represents our novel approach to the problem. The way we actually optimize the objective, however, is rather generic and can in principle be replaced by a different optimization procedure. Thus, we will introduce our method in two separate sections, of which the first is dedicated to the objective function and the second to the way we chose to optimize it.

A. Objective Function

Specifically, our method seeks spatial filters $\mathbf{w}_x$, $\mathbf{w}_y$, and a temporal filter $\mathbf{w}_\tau$ that invert the generative model. The non-linear step of band power computation is included in our method but it is applied to the estimated EEG sources directly and thereby avoids the pitfalls outlined in Section IV. In particular the filters shall maximize the covariance between the projected y-signal and the convolved power of the projected x-signal. We denote the projected signals by

$$\mathbf{w}_x^\top \mathbf{x} = \hat{s}_x \text{ and} \quad (6)$$

$$\mathbf{w}_y^\top \mathbf{y} = \hat{s}_y. \quad (7)$$

Since $\mathbf{y}$ varies on a slower time scale than $\mathbf{x}$, we will assume that the data can be chunked up into epochs with index e . The individual epochs should be long enough to allow for a reasonable estimate of spectral power of $\mathbf{x}$, yet short enough that $\mathbf{y}$ changes

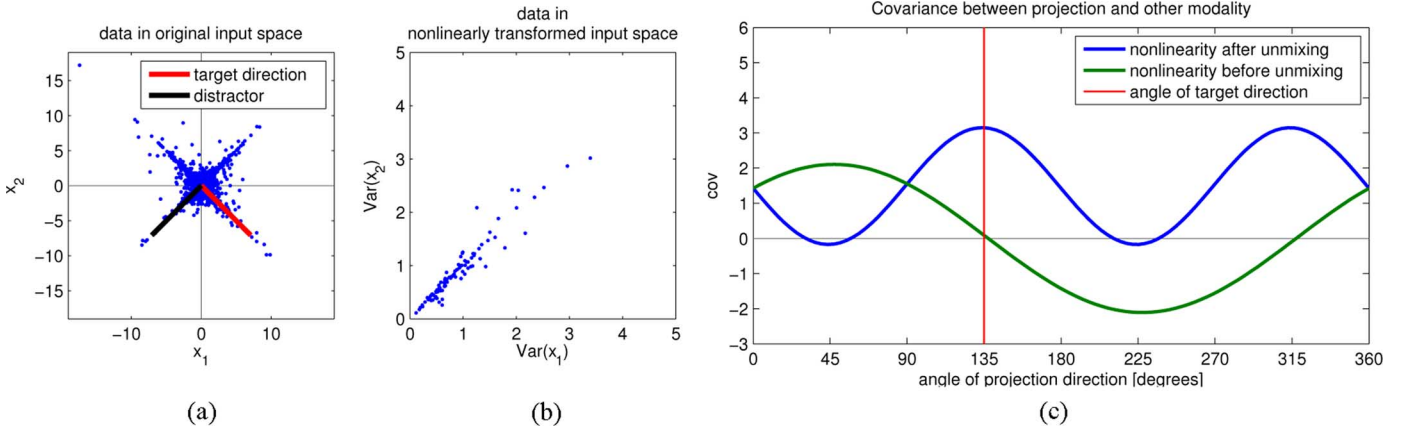


Fig. 3. Illustration for the limitations of channel-wise band power representations. *Left*: Scatter plot of data from two sources that have been linearly mixed. The original source directions are indicated with a red and black line. The band power of the source corresponding to the red direction (target source) is coupled to a variable z (not shown). *Middle*: The same data after applying a channel-wise nonlinearity (here variance as a proxy for band power). Note that information about polarity (sign) is lost. *Right*: Covariance between z and the band power of a projection of the data in the left scatter plot (blue line), and covariance between z and the projection of the data in the middle scatter plot (green line), both as a function of projection angle with respect to the x_2 axis. The vertical red line corresponds to the direction of the target source. In this simple example the maximum covariance obtained by unmixing the nonlinearly transformed data does not lead to the true source direction. Unmixing the data before applying the nonlinearity, however, allows to find the true source direction. See text for further explanation.

only very little within each individual epoch.² The epoch-wise power of $\hat{s}_x$ is estimated by squaring and averaging over the epoch, i.e.,

$$\begin{aligned} \hat{p}_{s_x}(e) &= \left\langle \left(\mathbf{w}_x^\top \mathbf{x}(t) \right)^2 \right\rangle_{t \in T_e} \\ &= \mathbf{w}_x^\top \mathbf{C}_{xx}(e) \mathbf{w}_x. \end{aligned} \quad (8)$$

In accordance with (2), we model the coupling function as a temporal filter on the power of $\hat{s}_x$. Thus we have

$$\hat{h}(\hat{p}_{s_x})(e) = \sum_n \mathbf{w}_{\tau n} \hat{p}_{s_x}(e - n), \quad (9)$$

where $\mathbf{w}_{\tau n}$ denotes the n th element of the vector $\mathbf{w}_\tau$ and N_τ the number of past epochs to be taken into account.

With the definitions given above, we can formulate the objective function that we want to maximize:

$$f_{\text{obj}}(\mathbf{w}_x, \mathbf{w}_y, \mathbf{w}_\tau) := \text{Cov} \left(\hat{h}(\hat{p}_{s_x}), \hat{s}_y \right), \quad (10)$$

with respect to the following norm constraints

$$\begin{aligned} \|\mathbf{w}_x\|_{\mathbf{C}_{xx}} &:= \mathbf{w}_x^\top \mathbf{C}_{xx} \mathbf{w}_x = 1, \\ \|\mathbf{w}_y\|_{\mathbf{C}_{yy}} &:= \mathbf{w}_y^\top \mathbf{C}_{yy} \mathbf{w}_y = 1, \text{ and} \\ \|\mathbf{w}_\tau\|_{\mathbf{B}} &:= \mathbf{w}_\tau^\top \mathbf{B} \mathbf{w}_\tau = 1, \end{aligned}$$

where $\mathbf{C}_{xx}$ and $\mathbf{C}_{yy}$ denote the with-signal covariance matrices and $\mathbf{B}$ denotes the auto-correlation matrix of $\hat{p}_{s_x}$ computed over all time lags that are considered. Together these constraints ensure that the objective function is bound between -1 and 1 .

²In most practical scenarios this setting is very natural, especially if $\mathbf{x}$ and $\mathbf{y}$ are acquired with different sampling rates. A typical example would be concurrent EEG and NIRS, where half a second represents an appropriate epoch length.

Without loss of generality, we assume $\hat{s}_y$ to have zero mean. We can then express f_{obj} in a concise way using tensor notation:

$$\begin{aligned} f_{\text{obj}} &= \left\langle \left(\hat{h}(\hat{p}_{s_x}) - \langle \hat{h}(\hat{p}_{s_x}) \rangle_e \right) (\hat{s}_y - \langle \hat{s}_y \rangle_e) \right\rangle_e \\ &= \left\langle \hat{h}(\hat{p}_{s_x}) \hat{s}_y \right\rangle_e - \underbrace{\left\langle \hat{h}(\hat{p}_{s_x}) \right\rangle_e \langle \hat{s}_y \rangle_e}_{=0} \\ &= \left\langle \sum_n \mathbf{w}_{\tau n} (\mathbf{w}_x^\top \mathbf{C}_{xx}(e - n) \mathbf{w}_x) (\mathbf{w}_y^\top \mathbf{y}(e)) \right\rangle_e \\ &= \sum_{n, i, j, k} \mathbf{w}_{\tau n} \mathbf{w}_{x i} \mathbf{w}_{x j} \mathbf{w}_{y k} \underbrace{\left\langle \mathbf{x}_i \mathbf{x}_j \right\rangle_{t \in T_{e-n}} \mathbf{y}_k(e)}_{:= (\mathbf{C}_{\mathbf{x}\mathbf{x}\mathbf{y}\tau})_{ijkn}} \\ &= \mathbf{C}_{\mathbf{x}\mathbf{x}\mathbf{y}\tau} \bar{\mathbf{x}}_{(1)} \mathbf{w}_x \bar{\mathbf{x}}_{(2)} \mathbf{w}_x \bar{\mathbf{x}}_{(3)} \mathbf{w}_y \bar{\mathbf{x}}_{(4)} \mathbf{w}_\tau, \end{aligned} \quad (11)$$

where $\langle \cdot \rangle_e$ denotes average over epochs, $\bar{\mathbf{x}}_{(i)}$ is the contraction of the i th dimension of an n -dimensional tensor via multiplication with a vector. For more information on $\bar{\mathbf{x}}_{(i)}$ and more tensor notation see [46].

Thus the optimization problem of maximizing the covariance between $\hat{h}(\hat{p}_{s_x})$ and $\hat{s}_y$ can be expressed as maximizing the n -mode vector product between the four-dimensional covariance tensor $\mathbf{C}_{\mathbf{x}\mathbf{x}\mathbf{y}\tau}$ and the vectors $\mathbf{w}_x$, $\mathbf{w}_y$, and $\mathbf{w}_\tau$. For convenience, we include the constraints directly in the objective function and thus arrive at

$$f_{\text{obj}}(\mathbf{w}_x, \mathbf{w}_y, \mathbf{w}_\tau) = \frac{\mathbf{C}_{\mathbf{x}\mathbf{x}\mathbf{y}\tau} \bar{\mathbf{x}}_{(1)} \mathbf{w}_x \bar{\mathbf{x}}_{(2)} \mathbf{w}_x \bar{\mathbf{x}}_{(3)} \mathbf{w}_y \bar{\mathbf{x}}_{(4)} \mathbf{w}_\tau}{\|\mathbf{w}_x\|_{\mathbf{C}_{xx}}^2 \|\mathbf{w}_y\|_{\mathbf{C}_{yy}} \|\mathbf{w}_\tau\|_{\mathbf{B}}}. \quad (12)$$

Our modeling approach (i.e., our objective function) allows for a meaningful interpretation of the results. Since we are seeking linear spatial filters $\mathbf{w}_x$ and $\mathbf{w}_y$, we are able to derive corresponding spatial patterns for each of the measurement modalities. By spatial patterns we refer to the columns of the mixing matrices $\mathbf{A}_x$ and $\mathbf{A}_y$ (see (1) and (3)). For a given

spatial filter $\mathbf{w}$ we compute the corresponding spatial pattern $\mathbf{a}$ according to

$$\mathbf{a} = \frac{1}{T} \sum_{t \in T} \mathbf{x}(t) (\mathbf{w}_x^\top \mathbf{x}(t)), \quad (13)$$

which represents the optimal³ linear prediction of the true source pattern [7]. The spatial patterns can be interpreted as the scalp projections of the sources whose activity is extracted via $\mathbf{w}_x$ and $\mathbf{w}_y$. In accordance with Blankertz *et al.* [47] we suggest to visualize the patterns rather than the filters in order to interpret the results.

B. Optimization Algorithm

In order to solve the maximization problem we chose an iterative procedure. Specifically we divide the original objective into three connected convex subproblems and solve those in turn until convergence. In order to arrive at the convex subproblems, we compute the first derivatives of f_{obj} with respect to $\mathbf{w}_x$, $\mathbf{w}_y$, and $\mathbf{w}_\tau$ and set them to zero. Here we present only the resulting equations for $\mathbf{w}_x$, $\mathbf{w}_y$, and $\mathbf{w}_\tau$.

The weight vectors that maximize our objective satisfy the following set of coupled equations

$$\mathbf{C}_{xx}^{-1} \frac{(\mathbf{C}_{xxy\tau} \bar{\mathbf{x}}_{(3)} \mathbf{w}_y \bar{\mathbf{x}}_{(4)} \mathbf{w}_\tau)}{\|\mathbf{w}_y\|_{\mathbf{C}_{yy}} \|\mathbf{w}_\tau\|_{\mathbf{B}}} \mathbf{w}_x = \lambda \mathbf{w}_x, \quad (14)$$

$$\mathbf{C}_{yy}^{-1} \frac{(\mathbf{C}_{xxy\tau} \bar{\mathbf{x}}_{(1)} \mathbf{w}_x \bar{\mathbf{x}}_{(2)} \mathbf{w}_x \bar{\mathbf{x}}_{(4)} \mathbf{w}_\tau)}{\|\mathbf{w}_x\|_{\mathbf{C}_{xx}}^2 \|\mathbf{w}_\tau\|_{\mathbf{B}}} = \lambda \mathbf{w}_y, \quad (15)$$

$$\mathbf{B}^{-1} \frac{(\mathbf{C}_{xxy\tau} \bar{\mathbf{x}}_{(1)} \mathbf{w}_x \bar{\mathbf{x}}_{(2)} \mathbf{w}_x \bar{\mathbf{x}}_{(3)} \mathbf{w}_y)}{\|\mathbf{w}_x\|_{\mathbf{C}_{xx}}^2 \|\mathbf{w}_y\|_{\mathbf{C}_{yy}}} = \lambda \mathbf{w}_\tau, \quad (16)$$

which hold simultaneously in stationary points of the objective function.

Please note that the (locally) optimal $\mathbf{w}_x$ is given as the solution to an eigenvalue problem given fixed $\mathbf{w}_y$ and $\mathbf{w}_\tau$. The (locally) optimal $\mathbf{w}_y$ is the result of contracting three dimensions of the covariance tensor via $\mathbf{w}_x$ and $\mathbf{w}_\tau$ and normalizing the resulting vector with the inverse of the covariance matrix of the y -signal. The (locally) optimal $\mathbf{w}_\tau$ is found by contracting three dimensions of the covariance tensor via $\mathbf{w}_x$ and $\mathbf{w}_y$ and normalizing the resulting vector with the inverse of the auto-correlation matrix. Thus, the solutions for $\mathbf{w}_x$, $\mathbf{w}_y$, and $\mathbf{w}_\tau$ depend on each other.

In our implementation we solve the three equations iteratively using the following steps:

- 1) Initialize $\mathbf{w}_x$, $\mathbf{w}_y$, and $\mathbf{w}_\tau$ randomly, compute $\mathbf{B}$ using $\mathbf{w}_x$.
- 2) Solve the eigenvalue problem in (14) and pick the eigenvector $\mathbf{w}_x$ corresponding to the largest eigenvalue.
- 3) Use $\mathbf{w}_x$ obtained in the previous steps to recompute $\mathbf{B}$.
- 4) Solve (15) using the current $\mathbf{w}_x$, $\mathbf{w}_\tau$, and $\mathbf{B}$ to obtain a new $\mathbf{w}_y$.
- 5) Solve (16) using the current $\mathbf{w}_x$, $\mathbf{w}_y$, and $\mathbf{B}$ to obtain a new $\mathbf{w}_\tau$.
- 6) Go back to step (1) and repeat until convergence.

Once the first set of corresponding components is found, the next set can be obtained by projecting the x - and y -signal into

the subspace orthogonal to $\mathbf{w}_x$ and $\mathbf{w}_y$ respectively and repeating the procedure outlined above. Criteria for convergence can be the difference between consecutive objective function values or a fixed number of iterations for example. In our implementation we use a mixture of both, i.e., we compute $f_{\text{obj}}(\mathbf{w}_x, \mathbf{w}_y, \mathbf{w}_\tau)$ after each iteration and stop if either the difference is smaller than some predefined threshold (we use 10^{-6}) or a maximum number of iterations is reached.

The actual runtime of the algorithm depends on a number of factors, such as the dimensionality of the input data (i.e., N_x , N_y , and N_τ), the number of epochs in the input stream N_e , and of course the convergence criteria. Assuming realistic values for the dimensionality of the problem ($N_x = 60$, $N_y = 50$, $N_\tau = 15$, and $N_e = 600$), our MATLAB implementation needs an average processing time of 0.1794 seconds (0.0522 seconds standard deviation) on a 3.6 GHz cluster node for one iteration of the procedure outlined above.

In practice we first employ a whitening and dimensionality reduction via PCA on both modalities separately. We reduce the number of dimensions such that at least 99% of the variance of the original data are preserved. Our optimization algorithm works greedily and is therefore not guaranteed to converge to a global optimum. Thus we choose to limit the number of iterations and also restart the optimization several times with different random initialization. For the simulations and real world data analysis shown in this paper, we have set the maximal number of iterations to 30 and restart the procedure 50 times, keeping only that set of weight vectors that yield the maximal objective function value.

VI. SIMULATIONS AND EXPERIMENTS

In this section we explain how we simulated data to test our method and give a description of real world data to which we apply our method.

A. Simulations

Artificial EEG and NIRS data is created using the following steps. Firstly, we generate time courses of 251 hypothetical band-limited EEG sources (1 target source and 250 background sources). For illustrative purposes we chose the α -band as the frequency band of interest, i.e., 8 to 12 Hz. The oscillatory signals are constructed such that the dynamics of their power modulation is slower than 1 Hz. This constitutes the EEG data in ‘source space’. Thereafter, physiologically plausible spatial patterns are generated via a realistic EEG forward model. Specifically, we place neural sources (i.e., electrical dipoles, here with randomly chosen orientation) at randomly chosen locations in 3D voxel space and compute the resulting scalp projections, which we denote with the vector $\mathbf{a}^i \in \mathbb{R}^{N_x}$ for the i th source. Using these scalp projections, we separately construct the sensor space representation of the target source (denoted by $\mathbf{x}_t$) and the sensor space representation of background neural activity (denoted by $\mathbf{x}_{\text{bg}}$):

$$\begin{aligned} \mathbf{x}_t(t) &= \mathbf{a}^1 s_1(t) \\ \mathbf{x}_{\text{bg}}(t) &= \sum_{i=2}^{251} \mathbf{a}^i s_i(t). \end{aligned}$$

³Optimal in a least squares sense

Additionally we add Gaussian distributed noise (zero mean and unit variance), which is spatially as well as temporally uncorrelated. The noise vector is denoted by $\epsilon(t)$. These three constituents (source signal, background activity and sensor noise) are individually normalized such that the corresponding full data matrices (e.g., $\mathbf{X}_t = [\mathbf{x}_t(1), \dots, \mathbf{x}_t(T)]$ and $\mathbf{X}_{bg}$ as well as $\mathbf{X}_\epsilon$ defined accordingly) have unit ‘global’ power, i.e., element-wise squaring of all the entries in $\mathbf{X}_t$ and averaging the result across time and channels yields 1. Thus, the signal and the two noise types have equal power before they are linearly combined to constitute the simulated EEG. Finally, the normalized matrices are combined according to parameterized equation

$$\mathbf{X} = \gamma \mathbf{X}_t + (1 - \gamma) \mathbf{X}_{bg} + \gamma_\epsilon \mathbf{X}_\epsilon. \quad (17)$$

The parameters γ and γ_ϵ control the relative weightings: γ_ϵ governs the ratio between the signals of the background sources and the sensor noise, while the signal-to-noise ratio is given by γ . The value of γ_ϵ is fixed to 0.1 for all simulations and the value of γ is varied. The number of channels of the simulated EEG is set to 64.

The simulated NIRS consists of 50 slow oscillatory background source signals and one target source signal that were created as follows. For the background sources, we filter white noise with a 1 Hz low-pass filter. Thereafter, we convolve the slowly varying noise with a hemodynamic response function (HRF) which was generated using the SPM package with default parameters. The target NIRS source signal is created by convolving the power time course of the target EEG source with the same HRF that was used for the background NIRS sources. NIRS source patterns are modeled as 2D Gaussian activity profiles. After patterns and time courses of the target and background NIRS are created, the artificial NIRS data is then projected to measurement space via the same SNR parameterized linear model that was used for the artificial EEG (see (17)).

We illustrate the performance and robustness of our method in comparison to CCA. We chose CCA as the best representative for a class of methods that yield linear projections and acts on channel-wise band power estimates. We consider CCA to be most appropriate contestant to mSPoC because it also acknowledges the multimodality of the data analysis problem, i.e., it assumes an underlying process that is reflected in both imaging modalities. Here we employ a version of CCA that is also able to account for a temporal filtering of the projection of the EEG power, namely convolutive CCA [33], [34].

In our simulations we vary the SNR (γ) of the artificial data. For a given value of γ we generate 20 minutes of data. mSPoC and CCA are trained on the first half of the data (i.e., the training data) and the resulting components ($\mathbf{w}_x$, $\mathbf{w}_y$, and $\mathbf{w}_z$) are then applied to the second half of the data (i.e., the test data). We measure the performance of the two methods as the correlation between the convolved power of the estimated EEG source and the time course of the estimated NIRS source, i.e., $\text{Corr}(\hat{p}_{s_x}, \hat{s}_y)$ (see (8) and (7)), on the test data. Thus, the reported correlations are obtained exclusively from the test data. This value indicates the ability of the methods to extract the coupled dy-

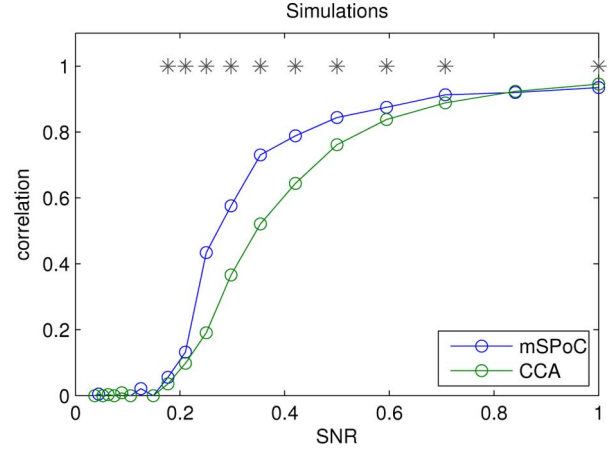


Fig. 4. Obtained correlations (medians over 250 repetitions) as a function of signal-to-noise ratio for simulated EEG and NIRS data. Black stars indicate significant differences in performance ($p < 0.01$).

namics in the data and generalize beyond the training data. The procedure is repeated 250 times at each SNR level, each time with newly generated data, leading to a distribution of correlation values for each value of γ . We compare the distributions using the Mann-Whitney U test statistic.

B. Experiments

The experimental data we apply mSPoC and CCA to stems from an experiment that featured simultaneous EEG and NIRS recordings. The study is described in detail in Fazli *et al.* [15]. In this section we briefly summarize the experimental design, data acquisition and preprocessing steps.

1) *Subjects and Experimental Design:* 14 healthy, right-handed volunteers participated in an experiment which consisted of 2 blocks of motor execution by means of hand gripping. Subjects were asked to close and open their hands with a frequency of 1 Hz for 4 seconds per trial followed by a break of 10.5 ± 1.5 s. An arrow on a display indicated whether to use their left or right hand. They performed 24 trials per condition and block in randomized order resulting in 48 trials per condition.

2) *Data Acquisition:* During the task EEG and NIRS was simultaneous recorded. With the NIRS-System (NIRSout 8–16, NIRx Medizintechnik GmbH, Germany) we measured 24 channels located mostly around the motor areas as well as frontal and parietal areas at $f_{\text{NIRS}} = 6.25$ Hz sampling frequency. EEG measurement (BrainAmp by Brain Products, Munich, Germany) was performed using 37 Ag/AgCl electrodes with a sampling rate of $f_{\text{EEG}} = 5$ kHz and downsampled to 100 Hz by means of a Chebyshev type II digital filter. The optical probes are constructed such that they fit into the ring of standard electrodes allowing measurements at the same location for both instrumentations. Sensors of both systems were integrated into a standard EEG cap, resulting in distances between light source and light detectors between 2–3 centimeters.

3) *Data Analysis:* Raw NIRS data was transformed to concentration changes of HbO and HbR with the modified Beer-Lambert law [48], [49] and band-pass filtered leaving frequencies between 0.025 and 0.25 Hz where the hemodynamic re-

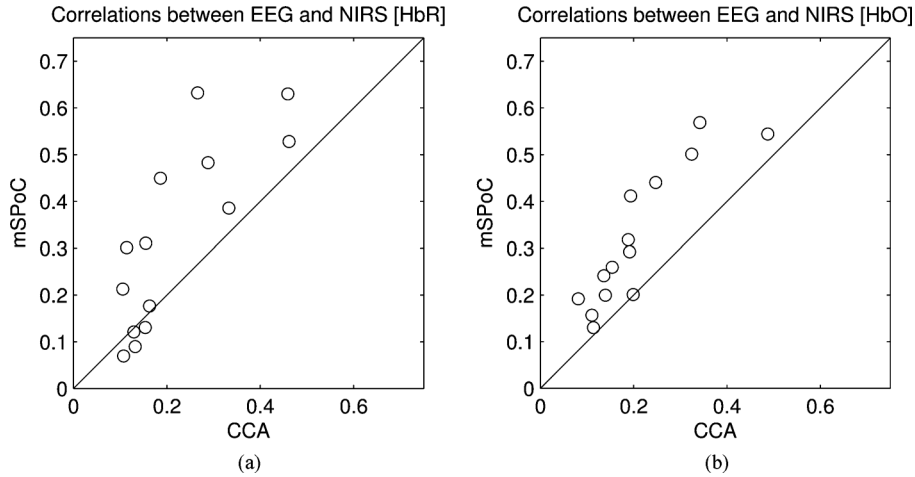


Fig. 5. Cross-validated correlations between EEG and NIRS (left HbR, right HbO) in the motor execution task for each subject. Results of mSPoC and CCA are compared. Each point corresponds to the correlations obtained for the first set of $\mathbf{w}_x$, $\mathbf{w}_y$, and $\mathbf{w}_\tau$ from a single subject by CCA (x -axis) and mSPoC (y -axis).

response is expected. EEG data was restricted to a subject specific frequency band that were used by Fazli *et al.* [15]. For the vast majority of subjects it was a narrow sub-band of the alpha or beta band range, which is known to reveal responses for motor execution [50]. Artifactual NIRS channels were excluded manually via visual inspection of the raw signals and the power spectrum. The band-pass filtered NIRS signals were sub-sampled such that their new sampling rate was 1 Hz.

Noisy EEG channels we identified and removed using a variance criterion. The EEG data was divided up into non-overlapping epochs of 1 second length. The chosen epoch length was not the result of an optimization procedure. Rather it was determined on the basis of a compromise between being long enough to contain a sufficient amount of data for within-epoch power/covariance estimation and being short enough to capture the dynamics of spectral modulations across epochs. For the mSPoC analysis we computed EEG covariance matrices within each epoch, while for the CCA analysis a channel- and epoch-wise Fourier transformation was conducted in order to extract the band power of interest. This yielded a time series of EEG covariance matrices for mSPoC and a multivariate power time series for CCA. After this procedure the temporal resolution of both input data sequences was reduced to 1 sample per second.

In order to avoid over-fitting and to test the generalization ability of the obtained mSPoC and CCA components we employ a chronological 5-fold cross-validation [51]. The input data is chronologically split up into 5 parts, which are called *folds*. The algorithms are then trained on 4 folds and the resulting components are applied to the remaining 5th fold. This procedure is repeated five times, each time with a different test fold. Finally, we report correlation values that are averaged across folds.

VII. RESULTS

A. Simulations

Fig. 4 shows the results obtained from the simulations in which we test the robustness of our approach with respect to

noise in the data. As can be expected, the obtained correlations of the extracted components falls off with a decrease in SNR (lower values of γ), meaning that mSPoC and CCA both fail to extract the underlying process that of course still present. However, we can see that mSPoC is en par with CCA in high SNR regimes and significantly outperforms CCA in medium to lower SNR regimes, suggesting that SPoC is much more robust with respect to noise.

We would like to point out, that the reported correlations are obtained on (unseen) test data. Therefore the correlations are likely to be smaller when compared to those on the training data. However, these values represent the generalization ability beyond training data. By definition, CCA obtains maximal correlations between two multivariate data sets, which is not guaranteed for mSPoC. However, the two algorithms work on different representations of the data and therefore mSPoC still has the possibility to outperform CCA.

The results of these simulations are promising with respect to real data, because the SNR in actual recordings is likely to be small.

B. Experiment

Fig. 5 shows the results of applying mSPoC and CCA to the simultaneously measured EEG and NIRS data of 14 subjects. We plot the cross-validated correlation values of the first set of components for each subject in a scatter plot. This way one can easily compare the performance of mSPoC (y -axis) and CCA (x -axis) for each subject (single point in the scatter plot). Each point above the diagonal indicates that higher correlations were found with mSPoC as compared to CCA.

The scatter plots in Fig. 5 indicate that mSPoC achieves higher correlations in both analysis settings, i.e., when the convolved band power of EEG components is correlated with NIRS components (HbR and HbO). In Fig. 6 we show the spatial patterns of the extracted components of mSPoC and CCA for a representative subset of subjects. The way we obtained the spatial patterns for the filters $\mathbf{w}_x$ and $\mathbf{w}_y$ is similar. The patterns for mSPoC were obtained as explained in Section V-A, (13) for EEG and NIRS. For CCA, NIRS patterns were also obtained

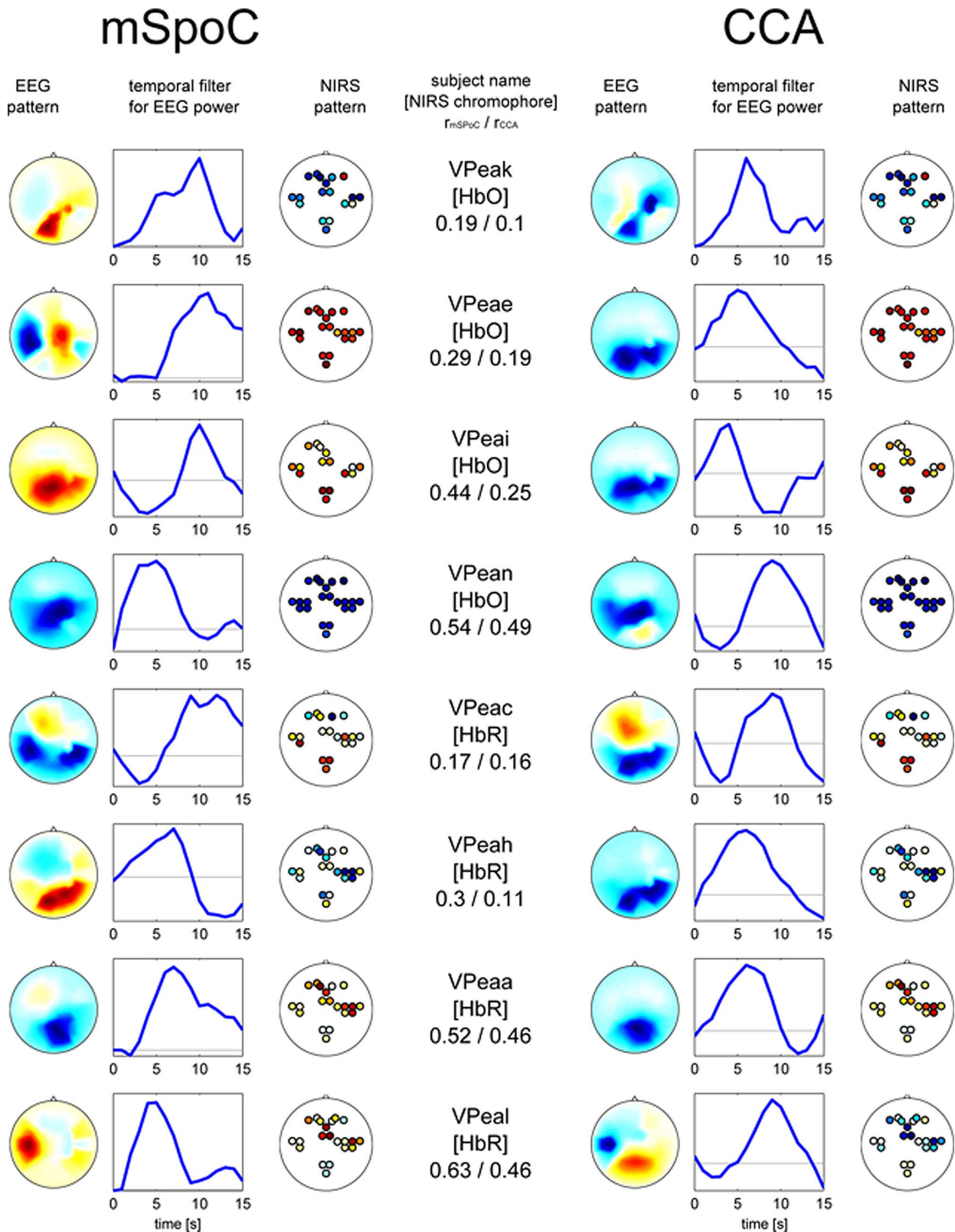


Fig. 6. Spatial patterns corresponding to $\mathbf{w}_x$ and $\mathbf{w}_y$, as well as the temporal filter for extracted EEG power $\mathbf{w}_t$. Each row shows the results obtained for a single subject, where on the left side we show the patterns obtained from mSpPoC and on the right side the patterns obtained from CCA. We chose 4 representative subjects for each NIRS chromophore but not the same subject twice, thus showing 8 different subjects in total (out of 14). In the middle column we show the subject code, the respective NIRS chromophore and the cross-validated correlation values between the convolved EEG component power time course and the time course of the NIRS component, as obtained by mSpPoC and CCA, respectively.

by means of (13). For the EEG band power patterns obtained via CCA, we used the time courses of the channel-wise EEG band power and not the EEG time-domain signals.

The majority of spatial patterns and temporal filters seem biologically plausible, in particular in those subjects where high correlations between the components could be obtained.

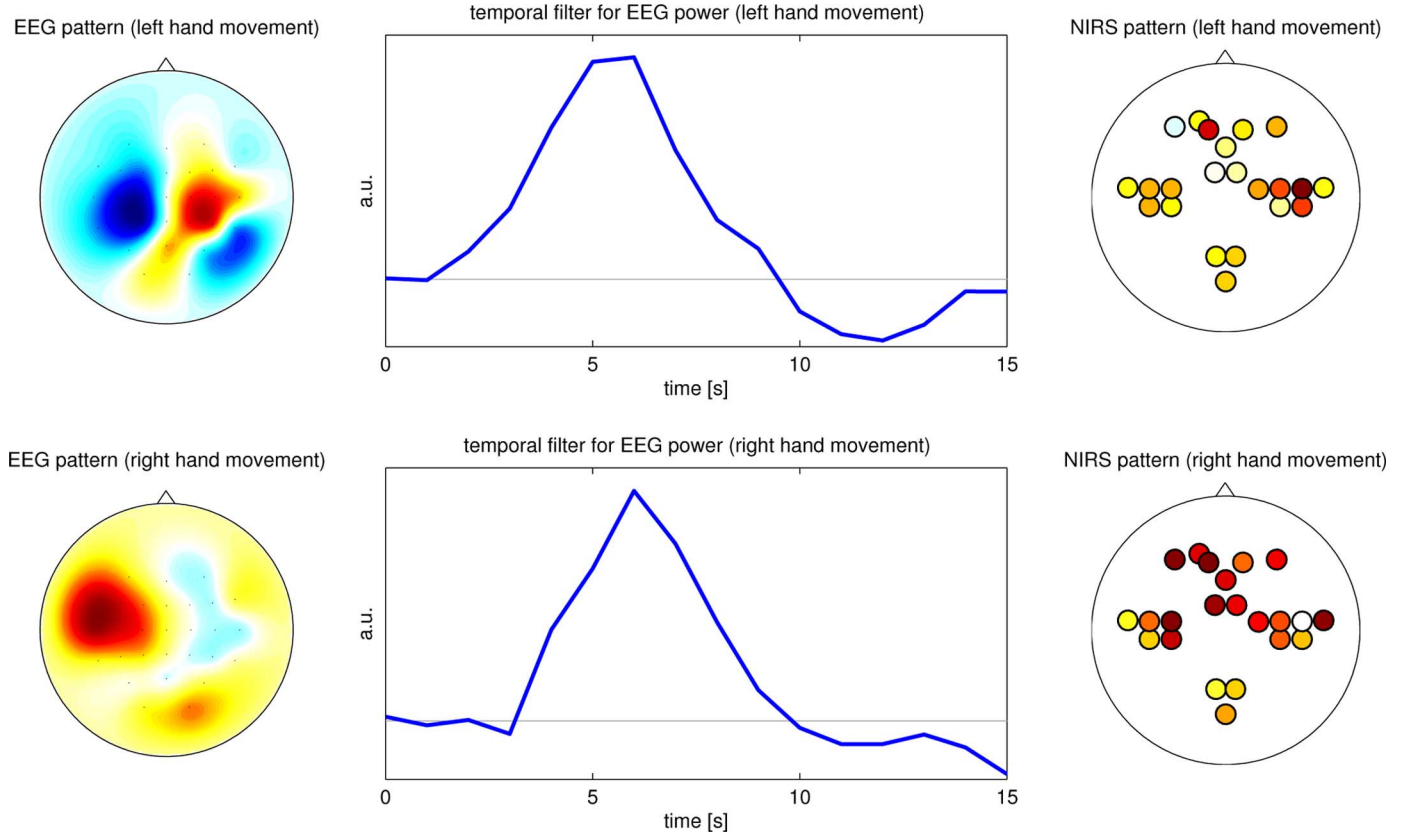


Fig. 7. Exemplary results for one subject (VPeau) as derived by mSPoC. The scalp-plots on the left side show the EEG pattern that corresponds to the obtained filter $\mathbf{w}_x$. In the middle plot we show the temporal filter for the EEG power of the component shown left. The rightmost scalp-plots depict the spatial pattern that corresponds to the filter $\mathbf{w}_y$, i.e., the NIRS patterns. The top row shows the results for applying mSPoC to left hand movement trials, while in the bottom row results for right hand movement trials are shown.

We find that mSPoC is able to extract motor-related sources from the two imaging modalities. This is illustrated in Fig. 7, which shows the EEG source patterns of one component set projected on the scalp and the temporal filter for one exemplary subject (VPeau) for each motor class separately. Plots in the top row are obtained from left hand movement trials and plots shown in the bottom row are obtained from right hand movement trials. The spatial pattern of the extracted EEG source in the top row of the figure is consistent with a dipole in motor-related areas of the right hemisphere. The NIRS pattern also has its largest weights located over motor areas on the right side of the scalp, which nicely reflects the neuro-physiology for left hand movements. The same is true for patterns obtained on the right hand movement trials. The EEG pattern is located over motor areas of the left hemisphere, which is consistent with the center of mass of the NIRS pattern. The shape of the derived temporal filter of both motor classes is in accordance with the HRF reported in the literature [10]. The filter functions realize a temporal low-pass filter with a time shift of 5 to 8 seconds.

VIII. CONCLUDING DISCUSSION

We presented a novel objective function for multimodal data integration that can be solved by readily available methods, e.g., the alternating least squares method exposed in Section V. The objective function is constructed to reflect the generative models of EEG and hemodynamic signals. The new analysis method

is called mSPoC and we compared it to a state-of-the-art competitor CCA. The results show that the performance of mSPoC exceeds that of CCA on simulated as well as real world data.

The proposed optimization scheme is efficient and fast enough to allow an application to real experimental data. However, we expect that there might still be room for improvement. There are a number of parameters that may be subjected to further optimization in order to reach better performance, for example the length of the epoch-window or the frequency band used to band-pass filter the EEG. Also the actual optimization algorithm may be improved by switching from our implementation to tensor decomposition techniques such as PARAFAC or CANDECOMP [46], [52]. There are other multi-way analysis approaches to either EEG power alone [53], [54] or to EEG power in conjunction to hemodynamics [36]. In these approaches, however, EEG band power is always computed in sensor space as part of the preprocessing and then added as a single dimension into the tensor or multi-way array. In our approach, the nonlinearity (i.e., estimation EEG band power) is directly incorporated into the objective function and thereby into the resulting tensor. This leads to an additional dimension in the tensor, but it is necessary to represent the EEG *source* power rather than the EEG *channel* power. Note that our approach is not the same as using a multi-way extension of CCA [55], because in our model the two first two dimensions of the tensor are deflated by the same weight vector ($\mathbf{w}_x$).

Our novel framework corresponds to a widely accepted generative model for the electro-physics of the EEG and the hemodynamics of neural sources [6], [7]. Therefore, our results are interpretable within this model. This is a particular difference to CCA- or PLS-based approaches for example. The spatial filters for the EEG band power obtained by our method are invertible to yield spatial patterns which correspond to the underlying sources. These EEG patterns can in principle be subjected to source localization algorithms (e.g., [56]–[58]), which is not true for patterns obtained from CCA or PLS, because these methods act on band power representations of the data.

We find physiologically plausible temporal coupling dynamics, which are in good accordance with previous studies [10], [34], [38]. Also the spatial patterns extracted by mSPoC are in line with established knowledge about the spatial location of the dipoles relevant for the experimental paradigm investigated here. A more thorough empirical validation with respect to the hemodynamic coupling between EEG and NIRS has been made possible with our method and is highly desirable. However, such an analysis is beyond the scope of this manuscript and is therefore subject to future work.

In the model for the vascular signal we assume the nonlinearity $p(\cdot)h(\cdot)$ to be known beforehand and the temporal HRF $h(\cdot)$ to be locally separable. A possible extension could be to also estimate different HRF for different neural sources simultaneously with the mixing matrices. An even more general extension might be to model the vascular signal as a fully non-separable spatio-temporal convolution of nonlinearly transformed source signals. In Biessman *et al.* 2012 [38] the authors find better correspondence between fMRI and intracranially measured neural signals using such non-separable spatiotemporal coupling functions. Their result is in line with previous findings about the variability of the HRF across brain regions, subjects [59] and cortical depth [60]. However, the improvement over separable models is generally rather small, so we restricted ourselves to the proposed model which yields a simpler and easier optimization problem. Note also, that our model already allows a certain class of non-separable hemodynamic responses, since it allows the shape of the temporal filter (i.e., the HRF) to be different for each mSPoC component. Thereby we implicitly assume *local* space-time separability within a brain region that constitutes a dipole, yet allow *global* space-time non-separability across the brain. As long as we do not have ultra-fine spatial resolution fMRI, we therefore do not expect our model to restrict the accuracy of the estimation. With further development of the instrumentation using high-density optical measures in combination with forward models [61], [62] and dipole localization procedures more advanced data integration methods will become more important. Data-driven methods are of particular importance for the formulation of more precise models of neurovascular coupling dynamics from non-invasive measurements of electrophysiological and hemodynamic data. Our method represents an important contribution towards data-driven multimodal data integration with a minimal set of assumptions. Future work includes the application of mSPoC to more real world data, a more thorough understanding of the analytic properties of the algorithm as well as further extensions to the underlying idea. For instance we plan to apply our method to concurrently recorded

EEG and fMRI. In order to deal with this kind of data, a kernel version of mSPoC will be developed to handle the problem of high dimensionality and comparably little training data. Also we plan to extend our objective to other kinds of nonlinearities. This will include, for example, higher moments of the data, such as kurtosis.

This paper has contributed a novel framework—mSPoC—that allows to address the following question: how can we correlate two raw multimodal multivariate signals that are spatially and temporally sampled differently and that are coupled through a nonlinear transformation. Our answer to this abstract question allows to find *unsupervised* projections, that discover unknown correlations between spectral power and a hemodynamic response. Thus, it has the potential to enable insights about complex cognitive processes in future studies.

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